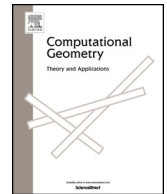




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Computing discrete harmonic differential forms in a given cohomology class using finite element exterior calculus

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ABSTRACT

Computational topology research of the past two decades has emphasized combinatorial techniques while numerical methods such as numerical linear algebra remain underutilized. While the combinatorial techniques have been very successful in diverse areas, for some applications, it is worth considering the numerical counterparts. We discuss one such application. Harmonic forms are elements of the kernel of the Hodge Laplacian operator and contain information about the topology of the manifold. If a particular cohomology class is chosen, the closed differential form with the smallest norm in that class is a harmonic form. We use these well-known facts to give an algorithm for solving the following problem: given a piecewise flat manifold simplicial complex (with or without boundary) and a closed piecewise polynomial differential form representing a cohomology class, find the discrete harmonic form in that cohomology class. We give a least squares based algorithm to solve this problem and show that the computed form satisfies the finite element exterior calculus (FEEC) equations for being a harmonic form. The piecewise polynomial spaces used are the spaces of trimmed polynomial forms, that is, arbitrary degree polynomial generalizations of Whitney forms which are used in FEEC. We also survey other methods for finding harmonic forms.

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1. Introduction

Most of the computational topology research of the past almost two decades or so has emphasized combinatorial techniques such as algorithms for computing persistent homology [1]. The continuing improvements in the efficiency and power of such techniques has resulted in a rapid adoption of computational and applied topology for applications in data science [2] and in fields as varied as complex materials [3], neuroscience [4], cosmology [5,6] and many others. In comparison with the combinatorial techniques, computational topology techniques based on numerical linear algebra remain underused. Numerical linear algebra has been a workhorse of the scientific computing community and many underlying computational libraries have been developed and optimized over the last 40–50 years. We demonstrate the use of such numerical techniques for computational topology in one application – the computation of a harmonic form in a given cohomology class.

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The harmonic form in a given cohomology class is an optimal cocycle: it is the differential form in that cohomology class which has the smallest L^2 norm. Thus the problem of finding a harmonic form in a given cohomology class is the cohomology analogue of the problem of finding an optimal cycle in a given homology class. The latter problem, which is referred to as the optimal homologous chain problem (OHCP) is known to be NP-hard when $\mathbb{Z}_2$ coefficients are used for homology. With integer coefficients, the first author (ANH) and collaborators proved various conditions characterizing when the OHCP problem can be solved in polynomial time using linear programming [7,8]. The use of $\mathbb{Z}_2$ or integer coefficients is natural in some applications of computational topology where one does not want to “smear” coefficients across a simplicial complex. Harmonic forms, on the other hand, arise in areas where real coefficients are more relevant and where such smearing is actually desirable. Cohomology and harmonic functions as well as forms arise naturally in partial differential equations [9,10], computer graphics [11–13], differential geometry and topology [14–17], and number theory [18]. In almost all of these fields, it is the real valued cohomology that is usually of relevance.

Solving the cohomologous analogue of OHCP has several important applications. For example, animators sometimes need to design a flow with specific characteristics, such as flowing in a particular way around rocks in a stream. For this purpose, vector fields corresponding to harmonic forms in a particular cohomology class can serve as the background vector field on which sources and sinks can be superposed. See [19] for a general introduction to design of vector fields in computer graphics. Another practical application of harmonic forms is in computational electromagnetism, in which the presence of conductors in a field in a cavity has the effect of introducing nontrivial homology. Solving Maxwell's equations in such instances sometimes requires specific cohomology classes and this can be provided by harmonic forms. For a recent paper in this area, see for instance [20]. The applications of the problem we pose and solve appear even in pure mathematics. For example, a currently open conjecture in number theory by Bergeron and Venkatesh requires the computation of the norm of harmonic 1-forms in specific cohomology classes on certain hyperbolic 3-manifolds. The first author ANH has been involved in a project with Nathan Dunfield [21] in which we are computing such harmonic forms using a method different from the one described in this paper. That method is specifically suited to hyperbolic manifolds.

We discuss numerical methods for finding discrete harmonic forms which are a piecewise polynomial approximation of the harmonic forms on simplicial manifold meshes. In particular, we wish to find the discrete harmonic form cohomologous to a given closed form representative of its cohomology class. That is, given an ω in the space of arbitrary degree piecewise polynomial differential forms such that $d\omega = 0$, we seek a discrete harmonic form u in the same space such that $u = \omega + d\alpha$ for some α . Here d is the restriction of the smooth exterior derivative operator to this space of piecewise polynomial forms. We can find such a cohomologous harmonic form by using a least squares method.

This is the content of our Theorem 1. The resulting linear system for the lowest order case can also be derived from the Hodge decomposition and in other equivalent ways. For the higher order polynomial approximation, the discrete system may also be obtained from the solution of the Hodge Laplacian problem using finite element exterior calculus (FEEC). The discrete harmonic form that we obtain satisfies the corresponding mixed formulation FEEC discretization of the Hodge Laplace's problem as in [9, Lemma 3.10].

Structure of the paper In Section 2 we recall some definitions and facts about harmonic forms and fields and the Hodge isomorphism theorem. Section 3 provides the relevant background for computational concepts needed. Section 4 contains our main result, Theorem 1 which is the key to our least squares algorithm for finding cohomologous harmonic forms. Section 5 gives an overview of related works, some of which are examined in greater detail in Section 6. Section 7 summarizes the results and conclusions.

2. Background on harmonic forms and fields

Let M be an oriented, compact n -dimensional Riemannian manifold, with or without boundary. The space of differential k -forms on M will be denoted $\Lambda^k(M)$. We first recall some facts about the isomorphism between cohomology and *harmonic forms* ($\ker \Delta$) or *harmonic fields* ($\ker d \cap \ker \delta$). (This material is based on [22]). The space of harmonic fields of degree k on M will be denoted $\mathcal{H}^k(M)$. For a *closed manifold* (compact manifold without boundary), harmonic forms and harmonic fields coincide, that is, $\ker \Delta = \ker d \cap \ker \delta$. However, in the case of compact manifolds with boundary ∂M , which we will refer to as ∂ -manifolds, we can only have that $\ker d \cap \ker \delta \subset \ker \Delta$ and there can exist harmonic forms which are not harmonic fields [23].

For closed manifolds there is an isomorphism between real valued cohomology and the space of harmonic forms. For compact ∂ -manifolds, even the space of harmonic fields is infinite dimensional due to the possibility of specifying boundary conditions. An isomorphism between harmonic fields and cohomology can be obtained by restricting harmonic fields by specifying certain boundary conditions.

Let $v_1, \dots, v_k \in T_p M$ for $p \in \partial M$. That is, $v_1, \dots, v_k$ are tangent vectors to M with base point on ∂M . Then the *tangential* component of a k -form $\omega \in \Lambda^k(M)$, denoted $\mathbf{t}\omega$, is defined by:

$$\mathbf{t}\omega(v_1, \dots, v_k) = \omega|_{\partial M}(v_1^\parallel, \dots, v_k^\parallel),$$

where $v_i = v_i^\parallel + v_i^\perp$ for $i = 1, \dots, k$ is an orthogonal decomposition with $v_i^\parallel$ being tangential to ∂M . The orthogonality is with respect to the Riemannian metric of M . The *normal* component of ω is defined by $\mathbf{n}\omega = \omega|_{\partial M} - \mathbf{t}\omega$. See [22, page 27] or [24, page 540]. It can be readily seen that:

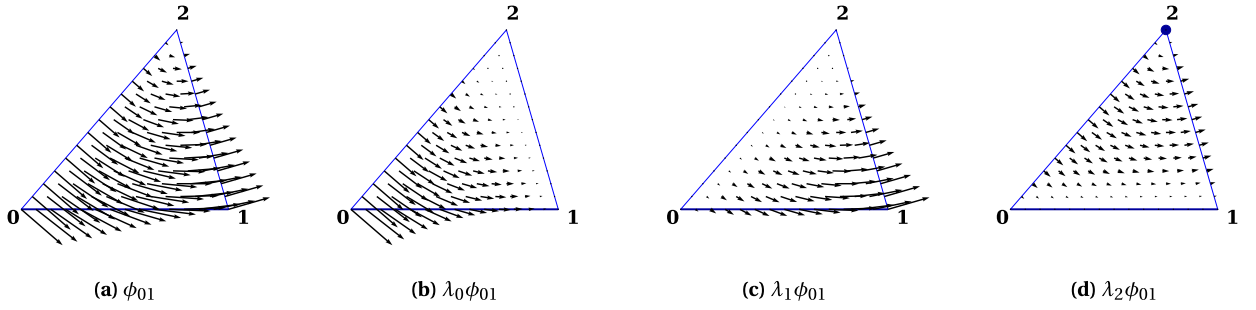


Fig. 1. Illustration of elements of trimmed polynomial differential forms spaces on a triangle T visualized as proxy vector fields. (a) A Whitney 1-form on T , i.e., an element of the space $\mathcal{P}_1^- \Lambda^1(T)$. This space has dimension 3 and the 1-form shown is a basis element corresponding to the bottom edge. There is one basis element corresponding to each edge. The other two basis elements are not shown. (b), (c), (d) Three of the basis elements of the space $\mathcal{P}_2^- \Lambda^1(T)$ for the triangle. This space has dimension 8. The forms shown in (b) and (c) correspond to the bottom edge. There are six such basis elements, two per edge. The form shown in (d) corresponds to the pair consisting of the bottom edge and opposite vertex. There are three such possible edge-vertex pairs of which only two are independent. Thus two are arbitrarily chosen to be basis elements.

$$\mathbf{n}\omega(v_1, \dots, v_k) = \omega|_{\partial M}(v_1^\perp, \dots, v_k^\perp).$$

A differential form ω is said to satisfy the *Neumann* or *absolute* boundary conditions if it has zero normal component ($\mathbf{n}\omega = 0$), and *Dirichlet* or *relative* boundary conditions if it has zero tangential component ($\mathbf{t}\omega = 0$). Let $\mathcal{H}_N^k(M)$ and $\mathcal{H}_D^k(M)$ denote the set of all harmonic fields satisfying the Neumann and Dirichlet boundary conditions, respectively. Then one has the following duality between analysis and topology.

Theorem (Hodge Isomorphism [22]). *If M is a closed manifold, $H^k(M; \mathbb{R}) \cong \mathcal{H}^k(M) = \ker \Delta_k$. If M is a compact ∂ -manifold, $H^k(M; \mathbb{R}) \cong \mathcal{H}_N^k(M)$ and $H^k(M, \partial M; \mathbb{R}) \cong \mathcal{H}_D^k(M)$.*

The space $H^k(M; \mathbb{R})$ is the (absolute) real k -cohomology vector space of M , and $H^k(M, \partial M; \mathbb{R})$ is the relative real k -cohomology vector space of M relative to its boundary. For ∂ -manifolds, we will only consider harmonic fields satisfying Neumann conditions. This is because the least squares method is based on a weak form of the Hodge Laplacian operator, and in that framework the Neumann conditions are automatically satisfied, that is they do not have to be enforced explicitly.

3. Computational background

3.1. Visualizing differential forms

In some cases differential forms can be visualized as vector fields as we have done in this paper. To understand these visualizations, note first that for any n -dimensional Riemannian manifold, spaces of 1-forms and vector fields are isomorphic. Specifically, for every differential 1-form α there is a unique vector field $\alpha^\sharp$ defined by $\langle \alpha^\sharp, V \rangle = \alpha(V)$ for all vector fields V , where $\langle \cdot, \cdot \rangle$ is the Riemannian inner product. This technique is used for all the surface and planar visualizations in this paper. Similarly for $(n-1)$ -forms β one can use the Hodge star to first turn β into the 1-form $*\beta$ and then use the above isomorphism. This latter technique is used for the 2-form visualization on the solid mesh in Fig. 7. These vector fields are called the proxy vector fields for the corresponding differential forms.

3.2. Trimmed polynomial differential forms spaces

Our Theorem 1 involves the use of degree r trimmed piecewise polynomial spaces for differential k -forms. These spaces are denoted $\mathcal{P}_r^- \Lambda^k$. For $r \geq 2, k > 0$, these spaces are a relatively recent development coming from finite element exterior calculus [9,10] and as such may not be widely known amongst this journal's readership. This section is a very brief introduction to these spaces. For degree $r = 1$ the space $\mathcal{P}_1^- \Lambda^k$ was invented by Hassler Whitney [25] and is now usually called the space of Whitney k -forms. In a simplex of dimension n , there is one Whitney k -form associated with each k -dimensional face of the simplex.

Fig. 1 shows some elements of the trimmed polynomial differential form spaces $\mathcal{P}_1^- \Lambda^1(T)$ and $\mathcal{P}_2^- \Lambda^1(T)$ on a triangle T , visualized as proxy vector fields. All of these forms are defined in terms of barycentric coordinates and their differentials. We will denote the barycentric coordinate corresponding to vertex i as λ_i . This is the affine function that takes the value 1 on vertex i and 0 on the other two vertices of T .

Fig. 1(a) shows the basis element of $\mathcal{P}_1^- \Lambda^1(T)$ corresponding to the edge $[0, 1]$ defined by $\phi_{01} := \lambda_0 d\lambda_1 - \lambda_1 d\lambda_0$. For the standard Euclidean inner product, the vector field proxy is $\phi_{01}^\sharp = \lambda_0 \nabla \lambda_1 - \lambda_1 \nabla \lambda_0$. The other two basis forms ϕ_{02} and ϕ_{12} in $\mathcal{P}_1^- \Lambda^1(T)$ are defined similarly and are not shown in Fig. 1. The proxy field $\phi_{01}^\sharp$ shown in Fig. 1(a) is normal to the other two edges of T . The reason for this follows readily from the formula for $\phi_{01}^\sharp$. Moreover, the pullbacks to a shared face

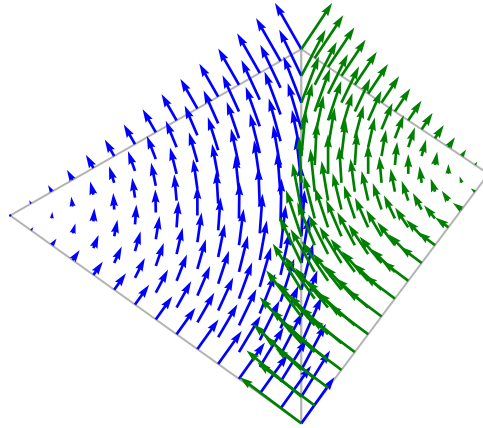


Fig. 2. Whitney 1-forms corresponding to a shared edge between two triangles. The tangential component of the proxy vector fields agree from both sides.

of basis forms in $\mathcal{P}_r^- \Lambda^k$ agree from both sides. For Whitney 1-forms this manifests itself as tangential continuity across shared faces. This is clearly visible in Fig. 2 which shows the Whitney 1-form corresponding to the shared edge between two triangles. This property has made the $\mathcal{P}_r^- \Lambda^k$ spaces very useful in finite element methods for solving partial differential equations, since this property provides the continuity for forms analogous to that provided by Lagrange basis elements for scalar functions. There are other very useful properties which are outside the scope of this brief introduction. See [26,9,10] for details.

An illustration of degree two basis elements is shown in Fig. 1(b)-(d). These are three of the basis elements of $\mathcal{P}_2^- \Lambda^1(T)$. The formulas for (b) and (c) are $\lambda_0 \phi_{01}$ and $\lambda_1 \phi_{01}$, each of which integrates to $1/2$ on the edge $[0, 1]$ and 0 on the other edges. Thus each edge yields two basis 1-forms. The two corresponding to edge $[0, 1]$ are shown in Fig. 1. The last basis 1-form shown in Fig. 1(d) is $\lambda_2 \phi_{01}$ and this is a 1-form corresponding to one edge-vertex pair, edge $[0, 1]$ and vertex 2. While there are three such pairs and three such 1-forms, only two such 1-forms are arbitrarily chosen, to complete a basis for $\mathcal{P}_2^- \Lambda^1(T)$. That is, a basis for $\mathcal{P}_2^- \Lambda^1(T)$ consists of six 1-forms like Fig. 1(b) and (c) and two like (d).

The space $\mathcal{P}_r^- \Lambda^k$ is called *trimmed* degree r because it is a proper subset of $\mathcal{P}_r \Lambda^k$, the space of degree r polynomial k -forms. The fact that these forms have the tangential continuity property is indicative of the fact that the spaces are trimmed, i.e., they do not contain all possible polynomial forms of degree r . Finally note the linear drop off of the vector field in Fig. 1(a) going from vertex 0 to 2 and vertex 1 to 2 along the two edges. This contrasts with the quadratic drop off going from vertex 0 to 2 in (b) and from 1 to 2 in (d). This is indicative of the polynomial degrees of the corresponding forms.

If Ω is a simplicial complex that is a piecewise linear approximation of a manifold M then each top dimensional simplex $T \in \Omega$ supports its own $\mathcal{P}_r^- \Lambda^k(T)$ and together, as a direct sum of these spaces, one obtains the global space $\mathcal{P}_r^- \Lambda^k(\Omega)$. By definition, the elements of this global space are piecewise smooth.

3.3. Combinatorial cochains and piecewise differential forms

For a simplicial complex Ω , the piecewise differential forms space $\mathcal{P}_r^- \Lambda^k(\Omega)$ is a finite dimensional vector space, and so its elements can be represented for purposes of computation as coefficients in a linear combination. In addition, for $r = 1$ there is a nice combinatorial interpretation in terms of what we will call combinatorial cochains. Let $C_k(\Omega; \mathbb{Z})$ be the group of integer-valued chains on Ω and $C^k(\Omega; \mathbb{R})$ be the dual space of cochains defined as $C^k(\Omega; \mathbb{R}) := \text{Hom}(C_k(\Omega; \mathbb{Z}), \mathbb{R})$, the space of homomorphisms from the chains to the additive group of reals. This is a vector space, and each k -simplex is the dual to and so corresponds to a basis cochain. Thus for top dimensional simplices T in Ω , each element of $\mathcal{P}_1^- \Lambda^k(T)$ can be represented by one number on each k -dimensional face of T . Consider for example, the edges $[0, 1]$, $[0, 2]$ and $[1, 2]$ of a triangle $[0, 1, 2]$. If the values assigned to these edges are a_0, a_1, a_2 then this combinatorial representation corresponds to the smooth differential form $a_0 \phi_{01} + a_1 \phi_{02} + a_2 \phi_{12}$ using the Whitney 1-forms corresponding to the edges. Let $W : C^k(T; \mathbb{R}) \rightarrow \mathcal{P}_1^- \Lambda^k(T)$, referred to as the Whitney map, be the interpolation map that takes these combinatorial cochains to Whitney k -forms. One important property of Whitney k -forms is the commuting property $dW = Wd$, where the d on the left is the exterior derivative and the d on the right is the coboundary operator. See [26] for details. Due to this property it is easy to start with a cocycle to create a closed form. For example, the red cocycles ω shown in Figs. 3, 4, 5 are elements of $C^k(\Omega; \mathbb{R})$ and $W\omega$ (which are not shown) would be smooth forms with support only in the triangles that contain the red edges.

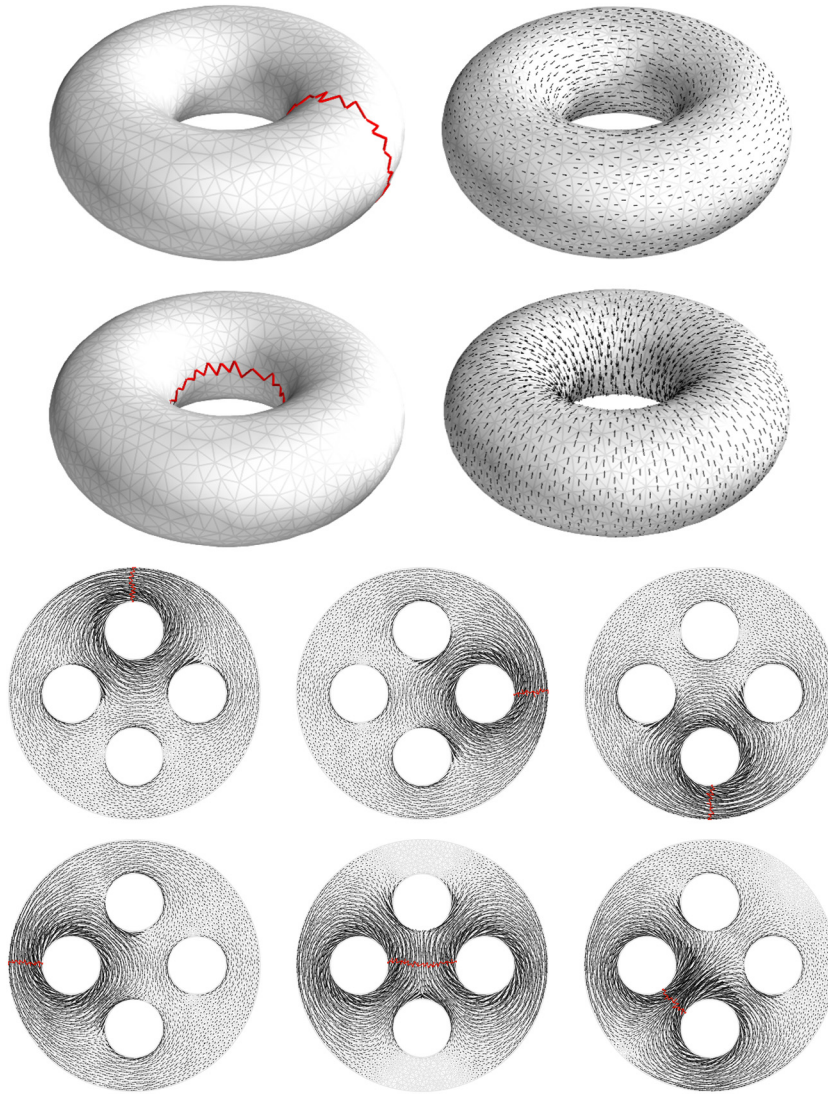


Fig. 3. Some example computations using the least squares method. *Top two rows* : Cocycles representing a cohomology basis for the torus are shown as red colored edges in the left figures. These are the ω cocycles of the text. The cocycles have value ± 1 on these edges and 0 on the other edges. The right figures show the proxy vector fields for the harmonic 1-forms in the corresponding cohomology classes. *Bottom two rows*: The nontrivial cocycles are marked in red. Note that the proxy vector fields for the harmonic Neumann 1-fields circulate *around* only those holes associated with the cocycle, and *past* others. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

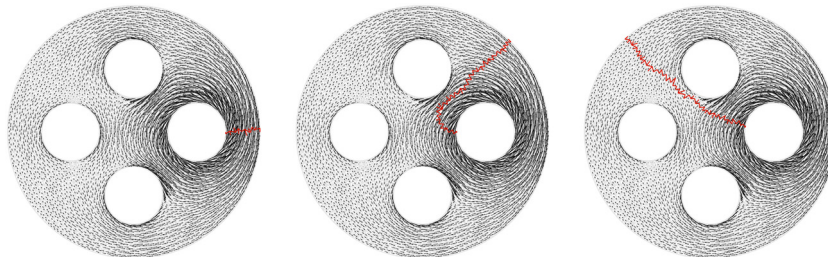


Fig. 4. Three different cocycles (ω of the text) representing the same cohomology class lead to the same harmonic Neumann 1-field when the least squares method is used.

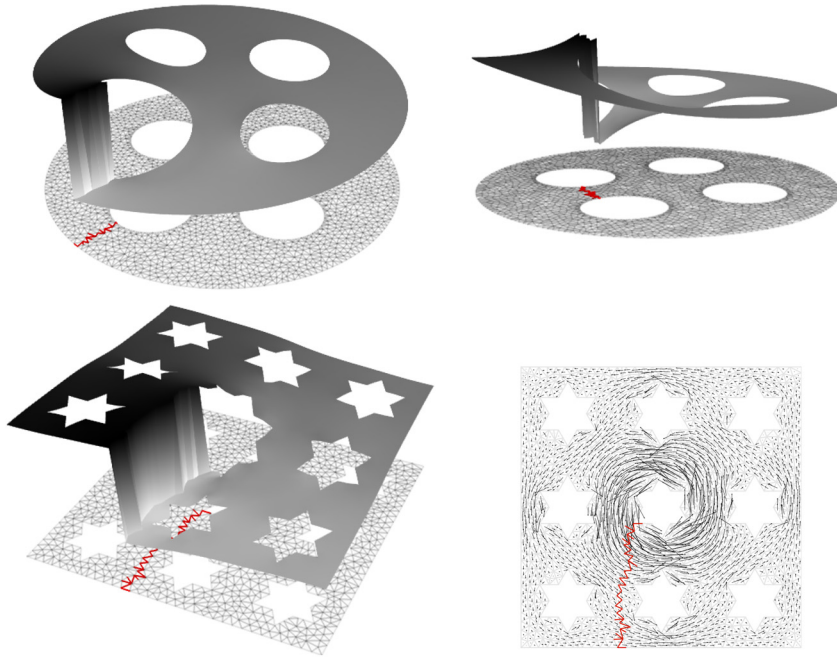


Fig. 5. Top row: Visualization of the 0-form α for two of the harmonic Neumann 1-fields. The 0-form α is visualized via surface elevation above the mesh and shading. Bottom row: Another example of a 0-form α and the corresponding harmonic Neumann 1-field but for a mesh with non-convex holes.

4. Least squares method

Let M be an n -dimensional Riemannian manifold which is approximated by a piecewise linear simplicial manifold complex Ω . The input to our algorithm is Ω . The precise details of the approximation of M by Ω are not relevant to what follows. For example, in computer graphics or numerical analysis application, Ω might be a piecewise linear surface triangulation embedded in $\mathbb{R}^3$. We will be using $\mathcal{P}_r^- \Lambda^k(\Omega)$ the spaces of piecewise, arbitrary order trimmed polynomial differential forms introduced in Section 3.2.

The proof of the Hodge isomorphism theorem quoted in Section 2 uses a calculus of variations argument to show that there is exactly one harmonic form in each cohomology class and that it is the L^2 -norm minimizer in that class. Motivated by this, we prove the following.

Theorem 1. Let $\omega \in \mathcal{P}_r^- \Lambda^k(\Omega)$ be a closed form. Let $\alpha \in \mathcal{P}_r^- \Lambda^{k-1}(\Omega)$ be a solution of:

$$\langle d\alpha, d\tau \rangle = -\langle \omega, d\tau \rangle \text{ for all } \tau \in \mathcal{P}_r^- \Lambda^{k-1}(\Omega). \quad (1)$$

Then the following are true.

- (i) A solution α of (1) exists and $d\alpha$ is unique,
- (ii) $u = \omega + d\alpha$ is a solution of the FEEC discretization of Hodge Laplace's equation. That is, u satisfies:

$$\begin{aligned} \langle d\tau, u \rangle &= 0 \quad \text{for all } \tau \in \mathcal{P}_r^- \Lambda^{k-1}(\Omega), \\ \langle du, dv \rangle &= 0 \quad \text{for all } v \in \mathcal{P}_r^- \Lambda^k(\Omega). \end{aligned} \quad (2)$$

The inner products are for piecewise polynomial differential forms in $\mathcal{P}_r^- \Lambda^{k-1}(\Omega)$ and $\mathcal{P}_r^- \Lambda^k(\Omega)$, and thus in $L^2 \Lambda^{k-1}(\Omega)$ and $L^2 \Lambda^k(\Omega)$, respectively.

- (iii) If $\partial\Omega \neq \emptyset$ then $\mathbf{n}u = 0$ on $\partial\Omega$.

Remark 4.1. The inner products on degree r polynomial k -forms used in the statement of Theorem 1 are all L^2 inner products. These are defined by integrating pointwise inner product of polynomial forms. (The pointwise inner product of polynomial forms is just the restriction of the smooth pointwise inner product to the polynomial subspaces.) For a smooth Riemannian manifold M , given $\alpha, \beta \in \Lambda^k(M)$ the L^2 inner product is $\langle \alpha, \beta \rangle := \int_M \alpha \cdot \beta \mu$ where the dot used inside the integral is the pointwise inner product of k -forms and μ is the volume form related to the Riemannian metric of M [24]. The pointwise inner product of k -forms can be built up from that of 1-forms with the inner product of the 1-forms simply

being that of the corresponding proxy vector fields. For details of the smooth case see [27], and for the lowest degree polynomial case see [28]. The equations (1) and (2) lead to linear systems by using basis elements of the polynomial spaces involved.

Remark 4.2. Since u satisfies the discrete Hodge Laplace's equation, we will refer to it as a discrete harmonic form if $\partial\Omega = \emptyset$ and discrete harmonic Neumann field if $\partial\Omega \neq \emptyset$.

Proof. Existence of α follows from the fact that $d\mathcal{P}_r^-\Lambda^{k-1}(\Omega) \subset \mathcal{P}_r^-\Lambda^k(\Omega)$ and uniqueness of $d\alpha$ follows from a least squares argument.

For $u = \omega + d\alpha$ satisfying (1), $\langle u, d\tau \rangle = \langle \omega + d\alpha, d\tau \rangle = 0$ for all $\tau \in \mathcal{P}_r^-\Lambda^{k-1}(\Omega)$. Also, $\langle du, dv \rangle = \langle d(\omega + d\alpha), dv \rangle = 0$ for all $v \in \mathcal{P}_r^-\Lambda^k(\Omega)$ since ω is closed. Thus, u satisfies (2).

For the first inner product in (2), the natural boundary condition is that $\text{tr} * u = 0$ [9, Equation (60)] where tr denotes the pullback j^* under the inclusion $j : \partial\Omega \hookrightarrow \Omega$. By equation (2.26) in [22], we have that “the tangential component $\mathbf{t} * u$ is uniquely determined by the pull back $\text{tr} * u$ ”. Thus, $\mathbf{t} * u = 0$. By equation (2.27) in [22, Proposition 1.2.6], $\mathbf{t} * u = *(\mathbf{n}u)$. Thus, $*(\mathbf{n}u) = 0$ and so $\mathbf{n}u = 0$ on $\partial\Omega$. $\square$

Extending the statements in Section 3.3 from a single simplex to the simplicial complex Ω , for the lowest degree case $r = 1$, the space $\mathcal{P}_1^-\Lambda^k(\Omega)$ is isomorphic to the space of k -cochains on Ω , $C^k(\Omega; \mathbb{R})$ [26]. The spaces $C^k(\Omega; \mathbb{R})$ for $k = 0, \dots, n$ form a cochain complex with the transpose of the boundary operators being the cochain maps which we will denote as $d^k : C^k(\Omega; \mathbb{R}) \rightarrow C^{k+1}(\Omega; \mathbb{R})$ or sometimes just d . The Whitney map W defined on individual simplices in Section 3.3 extends to $W : C^k(\Omega; \mathbb{R}) \rightarrow \mathcal{P}_1^-\Lambda^k(\Omega)$. This has a one sided inverse map $R : \mathcal{P}_1^-\Lambda^k(\Omega) \rightarrow C^k(\Omega; \mathbb{R})$ called the de Rham map [29]. For any $u \in \mathcal{P}_1^-\Lambda^k(\Omega)$, we have that $WRu = u$. We can now define the inner product of any two cochains $\hat{u}, \hat{v} \in C^k(\Omega)$ using $\langle W\hat{u}, W\hat{v} \rangle$ as $\hat{u}^T *_k \hat{v}$ where $*_k$ is the matrix representation of this inner product for the basis of $\mathcal{P}_1^-\Lambda^1(\Omega)$, and is referred to as the mass matrix.

For the lowest order case, (1) reduces to the weighted linear least squares problem:

$$\left(d^{k-1}\right)^T *_k d^{k-1} \hat{\alpha} = -\left(d^{k-1}\right)^T *_k \hat{\omega}, \quad (3)$$

where $\hat{\alpha} = R\alpha$ and $\hat{\omega} = R\omega$. This is the normal equation for the weighted least squares problem $d^{k-1} \hat{\alpha} \simeq -\hat{\omega}$. The matrix $\left(d^{k-1}\right)^T *_k d^{k-1}$ may have a nontrivial kernel. For example, for $k = 1$, $\ker d^0$ will have dimension equal to number of connected components in the complex. Note that (3) is equivalent to $\delta^k d^{k-1} \alpha = -\delta^k \omega$ which is $\delta^k(\omega + d^{k-1} \alpha) = 0$ where $\delta^k := \pm *_{n-k+1} d^{n-k} *_k$ is the discrete codifferential. This should make the connection to $\omega + d^{k-1} \alpha$ being harmonic more evident since we also have that $d^k(\omega + d^{k-1} \alpha) = 0$.

Remark 4.3. The harmonic form u cohomologous to a given closed form ω in a cohomology class can also be obtained by considering the Hodge decomposition of ω . For a general L^2 k -form ω , the Hodge decomposition is the unique decomposition $\omega = d\alpha + \delta\beta + u$ where $\alpha \in L^2\Lambda^{k-1}$, $\beta \in L^2\Lambda^{k+1}$ and $u \in L^2\Lambda^k$ is the harmonic k -form. Thus for a closed k -form ω cohomologous to harmonic form u , $\delta\beta = 0$ in the Hodge decomposition. In the finite dimensional case where we seek polynomial differential forms, this leads to solving for u by first obtaining α from the least squares problem $d\alpha \simeq \omega$.

Remark 4.4. Our Theorem 1 can be extended to the slightly more general case for a weakly closed ω . This will involve using a weak exterior derivative. Consequently, by extending the notion of being cohomologous to instead being weakly cohomologous in a natural way, our result will continue to hold.

Remark 4.5. In (3), the $*_k$ can also be replaced by the diagonal Hodge star of discrete exterior calculus [30].

FEEC has a well-developed and sophisticated theory of convergence and stability. As a corollary of our theorem which connects our least squares formulation to a FEEC formulation, we obtain the following optimal order convergence result.

Corollary 1. If u is a smooth harmonic form on M and $u_h = \omega + d\alpha$ where α is a solution of (1) then the L^2 error $\|u - u_h\| = O(h^r)$ where $h \rightarrow 0$ parametrizes a sequence of shape regular simplicial meshes.

Here h is a characteristic length associated with the simplicial mesh (such as the maximum edge length) and shape regularity is a standard assumption in finite element analysis needed for convergence statements to hold [31]. Shape regular meshes avoid pathological simplicial elements such as very thin or very flat triangles or tetrahedra etc.

Proof. This follows from [9, Theorem 3.9, Lemma 3.10, Section 3.5, Section 5.4] as a consequence of our Theorem 1. $\square$

4.1. Numerical results

All computations in this paper were done using the Python language with SciPy, NumPy, and PyDEC [32] packages. The top two rows of Fig. 3 show the harmonic cochains cohomologous to given nontrivial cocycles on a torus surface. The bottom two rows of Fig. 3 show several examples on a planar mesh with holes. To single out a particular hole, so that the harmonic cochain proxy vector field will circulate around that hole, one picks a cocycle connecting that boundary to the outer boundary. Connecting two holes results in a harmonic cochain that circulates about those two holes. For the cochains shown in the third row of Fig. 3, from left to right, the values of $\|\Delta u\|_{*1}$ relative to $\|u\|_{*1}$ are approximately 4.12×10^{-11} , 7.32×10^{-11} and 3.02×10^{-11} , respectively. Similarly, for the cochains in the bottom row, from left to right, these values are 4.98×10^{-11} , 5.73×10^{-11} and 5.64×10^{-11} , respectively. Fig. 4 shows that the least squares method (as expected) finds the same harmonic cochain when very different initial cocycles from the same cohomology class are given as input. If the cohomologous cochains are denoted u, u', u'' from left to right, respectively, then the differences between them are $\|u - u'\|_{*1} = 1.1 \times 10^{-14}$, $\|u - u''\|_{*1} = 2.8 \times 10^{-14}$ and $\|u' - u''\|_{*1} = 2.2 \times 10^{-14}$.

Fig. 5 illustrates the scalar function α found by solving (1) for the lowest order case $\mathcal{P}_1^- \Lambda^0(\Omega)$. In vector calculus terminology, its unique gradient $d\alpha$ is added to the cohomology basis representative ω to obtain the harmonic Neumann 1-field. The figure also shows an example with non-convex holes.

4.2. Linear solvers for the least squares method

The normal equations corresponding to our least squares system (1) are positive semidefinite. This is more clearly apparent when these normal equations are formulated for the lowest degree spaces of Whitney forms. Then the matrix $(d^{k-1})^T *_k d^{k-1}$ in the normal equation system (3) is positive semidefinite since d^{k-1} will typically have a nontrivial kernel. For example, for $k = 1$ for a connected domain, the space of constant functions on the domain is in the kernel of d_0 . In this case, it is easy to make the system nonsingular (mod out the nontrivial kernel) by fixing the value at a vertex and adjusting the linear system accordingly. For the case of 2-cochains in tetrahedral meshes however, the kernel of d_1 can be large. Let Ω be a three-dimensional manifold simplicial complex. Simple linear algebra and elementary topology reveals that the $\dim(\ker d_1) \geq N_0 - \chi(\Omega)$ where N_0 is the number of vertices and $\chi(\Omega)$ is the Euler number (the alternating sum of Betti numbers at all dimensions) [33]. For example, for a connected domain with boundary, we will have $\dim(\ker d_1) \geq \text{number of vertices} - 1 + \text{number of solid handles} - \text{number of cavities}$. By refining the mesh this kernel dimension can be made arbitrarily large. If a direct solver is to be used for solving (3) then one must mod out this potentially large nontrivial kernel. This can be done by finding a basis for the kernel of the d_{k-1} using standard linear algebra techniques and then solving a constrained least squares problem by looking for an α orthogonal to the kernel. An alternative is to use iterative Krylov solvers as they work well even in the presence of a nontrivial kernel and this is the approach we chose in our experiments. Specifically, we used a conjugate gradient solver without any preconditioning or modifications. Algebraic multigrid is another very efficient alternative whose effectiveness for this problem has been shown in [34].

4.3. Finding the initial nontrivial cocycles

In this paper we assume that a nontrivial cocycle is given. Our aim here is not to give algorithms for finding a cocycle. However, a few words about this are in order. An initial nontrivial cocycle in a cohomology class can be found in a number of ways. For surfaces, efficient algorithms to do this exist. Using annotations for edges as described in [2] (see question 7, Chapter 4 as well as Proposition 4.5), in $O(gn)$ time where g is the genus of the two dimensional simplicial complex and n is the number of simplices, one can find a homology basis for the topological dual (for example, barycentric dual) graph of the triangulation. One can then use Poincaré-Lefschetz duality [33] to get a cohomology basis on the primal mesh. For a boundaryless manifold simplicial complex, one would start with nontrivial cycles on the dual graph. But in case of a manifold with boundary, due to Lefschetz duality, one has to start with a nontrivial relative cycle on the dual mesh, relative to the boundary. One can also start with a random cochain and compute the desired nontrivial cocycle using a Hodge decomposition with standard inner product [34].

5. Overview of related works

The discussion in this section is for $\mathcal{P}_r^- \Lambda^k(\Omega)$ for $r = 1$ but extends to general $r \geq 2$. Cohomologous harmonic cochains can also be computed by first computing a harmonic cochain basis followed by some post processing. Such a basis can be obtained as eigenvectors of the zero eigenvalue of a discrete Δ_k . The problem of finding eigenvectors can be formulated (in the terminology of finite element methods) using a weak mixed or weak direct method.

For the weak mixed eigenvector method, consider the linear system for the unknowns σ and u :

$$\begin{aligned} \langle \sigma, \tau \rangle - \langle d^{k-1} \tau, u \rangle &= 0, \\ \langle d^{k-1} \sigma, v \rangle + \langle d^k u, d^k v \rangle &= 0, \end{aligned}$$

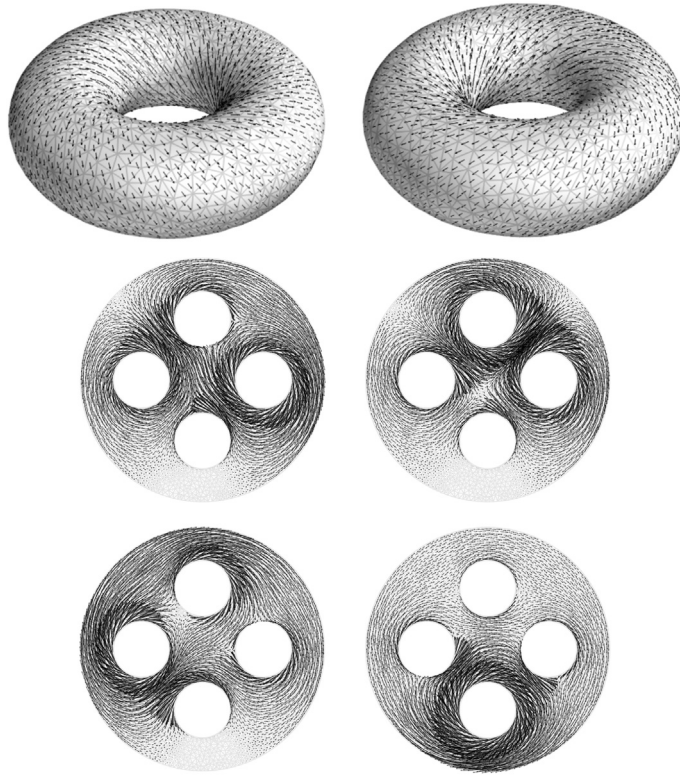


Fig. 6. Harmonic 1-fields produced by the mixed eigenvector method. The torus has a 2-dimensional space of harmonic 1-forms and the four-holed disc has a 4-dimensional space of harmonic Neumann 1-fields.

for all τ and v . Then (σ, u) is a solution if and only if $\sigma = 0$ and u is a harmonic k -form [9, Lemma 3.10]. For $\mathcal{P}_1^- \Lambda^k(\Omega)$, we discretize these equations and obtain the system matrix:

$$\begin{bmatrix} - *_{k-1} & (d^{k-1})^T *_{k-1} \\ *_{k-1} d^{k-1} & (d^k)^T *_{k+1} d^k \end{bmatrix}, \quad (4)$$

whose eigenvectors corresponding to the zero eigenvalue we seek. Fig. 6 shows results of the eigenvector calculations. The eigenvector methods will often suffice, if all that is needed is *some* harmonic basis, which may be the case as in finite element exterior calculus. Applications like vector field design in computer graphics may require more control over the process, namely being in a given cohomology class.

If a harmonic cochain basis is available, then an orthogonal projection to the harmonics can be used to obtain a harmonic cochain u cohomologous to a given cocycle ω .

For vector field design in computer graphics or in physical applications, the usual cases are dimension 2 with 1-cochains and dimension 3 with 1-cochains or 2-cochains. In the latter case, only solid handles and cavities are relevant since general 3-manifolds are typically not used in such applications. In all such cases, homology basis elements can be put in correspondence with topological features. Such a homology basis can be used by pairing with cohomology to find cohomologous harmonic cochains. However, this would require not only the entire harmonic cochain basis but also a full homology basis.

In [19] Fisher et al. introduced a method for designing vector fields on surfaces. The method allows one to design a 1-cochain (and hence a vector field) on a simplicial surface. The inputs are the sources and sinks (gradient part of the vector field), rotations (curl part of the vector field), as well as arbitrary linear constraints on the value that the cochain should take on edges. The computation then yields a 1-cochain satisfying the requirements in a weighted least squares sense. Gu and Yau [12] have a method that solves a Poisson-like equation at a different dimension and which also appears in their earlier work. The formulation is simple and straightforward but it leads to inefficient methods on general simplicial meshes. This method was further simplified by Desbrun et al. [13] who solve a Poisson's-like equation at a different dimension. The resulting linear systems in both methods suffer from numerical and scalability issues for general simplicial meshes.

6. Comparisons with other methods

The least squares method of Fisher et al. [19] mentioned in the previous section does not allow for cohomology constraints and so we will not discuss that method further. The first relevant method to compare with is that of Gu and Yau.

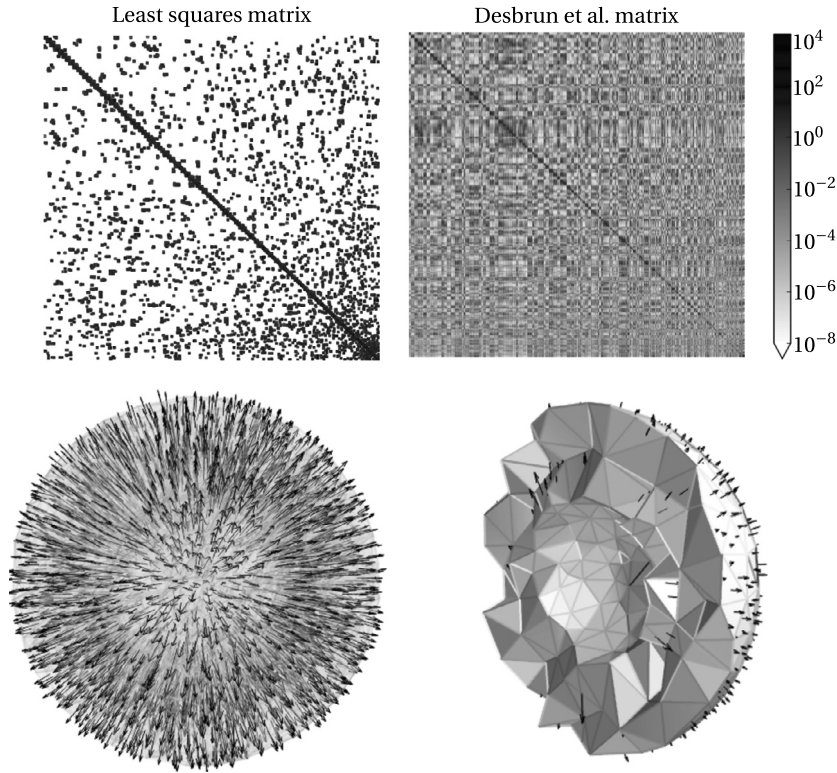


Fig. 7. Top: Magnitudes of nonzeros in operators using $\mathcal{P}_1^- \Lambda^1(\Omega)$. The least squares matrix (left column) is significantly sparser than the Desbrun et al. matrix (right column). This is due to the extra term in the Desbrun et al. matrix. The colorbar shows the magnitude of the nonzero components. The two matrices are of equal size, and are for finding harmonic Neumann 2-fields on the tetrahedral mesh of the solid annulus. Bottom: Proxy vector field of the harmonic Neumann 2-field on a tetrahedral mesh of the solid annulus (solid ball with cavity) using the least squares method (left column) and a cutaway view of this vector field showing the tetrahedral mesh (643 vertices and 5952 elements) for this problem (right column).

They start with a nontrivial cocycle ω representing a cohomology class in $H^k(X)$ and seek a cochain $\omega + d\alpha'$ such that $\Delta(\omega + d\alpha') = 0$. This leads to the linear system:

$$d^{k-1} *_k^{-1} \left(d^{k-1} \right)^T *_k d^{k-1} \alpha' = -d^{k-1} *_k^{-1} \left(d^{k-1} \right)^T *_k \omega.$$

The presence of the inverse Hodge stars in this system lead to numerical disadvantages.

Desbrun et al. solve a different Poisson's equation:

$$(d^{k-2} \delta^{k-1} + \delta^k d^{k-1}) \alpha' = -\delta^k \omega.$$

A solution α' to the above equation yields an $\omega + d\alpha'$ that is harmonic in the sense of this paper. Of course, if harmonic 1-cochains are being sought, then α' is a 0-cochain and δ^0 is the 0 operator. Thus the $d\delta$ term is not present. However, the $d\delta$ term is superfluous at every dimension as we have shown. Thus their linear system has an extra, unnecessary term. This extra term causes numerical and scalability problems.

In Fig. 7, we also compare the sparsity of the least squares and Desbrun et al. matrices for finding harmonic 2-cochains on a tetrahedral mesh of a solid annulus (a solid ball with an internal cavity). Both matrices are of the same size but the Desbrun et al. matrix is denser with 14.3 million nonzero entries versus 56 thousand nonzeros for our matrix. Here the increased density is due to the extra term in the Desbrun et al. system. The figure also shows the proxy vector for the computed harmonic Neumann 1-field using the least squares method and a cutaway view of the tetrahedral mesh on which this computation was performed.

The superior sparsity of the linear system matrix in the least squares method leads to improved solution time. To illustrate this, we compare the time taken for again finding harmonic 2-cochains on a tetrahedral mesh of a solid annulus. For the least squares method, using conjugate gradient method (without preconditioning), the time was 0.1181 seconds. For the Desbrun et al. method, this time was 1746 seconds. We also used a sparse solver in SuperLU for Desbrun et al. system and in this case, the time was 13.05 seconds. All times are averaged over many trials. Also, it may be possible to improve the times for both the methods by using preconditioners or special solvers.

7. Conclusions

We prove an isomorphism theorem that yields an algorithm for computing a harmonic form in a particular cohomology class. Our formulation and algorithm is in the space of trimmed piecewise defined degree r polynomial differential forms spaces $\mathcal{P}_r^- \Lambda^k$ that are a relatively recent development coming from the numerical analysis field of finite element exterior calculus (FEEC). The cohomology class is specified by selecting a closed form representative of the class. Since harmonic forms have the smallest L^2 norm amongst all the closed forms in the class, the problem we solve is an optimization problem. This is the real coefficient cohomologous dual of the optimal homologous chain problem in homology optimization. In our context real coefficients are relevant because harmonic forms and fields are smooth and thus spread out across the domain even if the cohomology representative is localized. To the best of our knowledge ours is the first work to solve the cohomologous harmonic form problem in spaces of arbitrary degree polynomial forms. This will permit the use of fewer, higher order elements when compared with lower order formulations. All other known methods for solving this problem use the lowest degree 1 trimmed polynomial spaces called Whitney forms. Even when compared at degree 1, our method is superior to all other known methods in terms of the size and sparsity of linear systems and resulting efficiency of solving these linear systems. As a corollary of our theorem, using the powerful convergence analysis of FEEC our computed harmonic forms converge to the smooth one in optimal order of convergence. To the best of our knowledge, this is the only convergence result amongst all the known methods for computing cohomologous harmonic forms. Relating our computation to FEEC has other advantages besides being able to import convergence results. It also implies that our results continue to hold for other types of elements in FEEC, such as serendipity elements. Our algorithm can also piggyback on future developments in FEEC, for example, development of elements for polygonal elements, developments of preconditioners for linear solvers and other improvements in linear solvers for the linear systems arising from FEEC equations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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